Note

- If $A \subseteq B$ and $B \subseteq A$, then $A = B$.
  In fact this is how we defined equality of sets.
- Empty set is a subset of every set.
  This is not easy to see ! Let A be any set. The only way for the empty set to be not a subset of A would be to have an element x in it but with x not in A. But how can x be in the empty set ? That is impossible. So this only way being impossible, the empty set must be a subset of A. (Is your head spinning? Think calmly, explain it to a friend, and you will agree it is alright !)
- Every set is a subset of itself. (Try and argue why?)

1.4.9. Proper Subset

Let A and B be two sets. If A is a subset of B and $A \neq B$, then A is called a proper subset of B and we write $A \subset B$.

For example,

If $A=\{1,2,5\}$ and $B=\{1,2,3,4,5\}$ then A is a proper subset of B ie. $A \subset B$.

1.4.10 Disjoint Sets

Two sets A and B are said to be disjoint if they do not have common elements.

In other words, if $A \cap B = \varnothing$, then A and B are said to be disjoint sets.

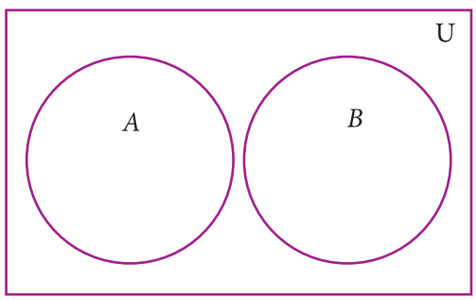


Fig. 1.3

Example 1.8

Verify whether $A=\{20, 22, 23, 24\}$ and $B=\{25, 30, 40, 45\}$ are disjoint sets.

Solution

$$A = \{20,22, 23, 24\} , B=\{25, 30, 40, 45\}$$

$$A \cap B = \{20,22, 23, 24\} \cap \{25, 30, 40, 45\}$$

$$= \{\ \}$$

Since $A \cap B = \varnothing$, A and B are disjoint sets.

Note

If $A \cap B \neq \varnothing$, then A and B are said to be overlapping sets .Thus if two sets have atleast one common element, they are called overlapping sets.

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